

Matthew Cromer

Education

University of Connecticut – *B.S. in Mechanical Engineering - 3.6 GPA*

Expected Graduation May 2027

- Dean's List Spring 2026 with a 3.9 GPA
- Pursuing a double concentration in Dynamic Systems and Control (Robotics) as well as Design and Manufacturing

Experience

Undergraduate Researcher, LINKS Lab - UConn

September 2026 – Present

- Working with Dr. Shalabh Gupta researching cooperative motion planning and navigation algorithms for swarm robotic systems.

Student Lab Assistant, Physics Department – UConn

August 2026 – September 2026

- Set up, tested, and maintained experimental equipment for the Physics Teaching Lab, troubleshooting instrumentation and managing inventory alongside teaching staff.

Mechanical Engineering Intern, Marmon Utility – Amherst, NH

May 2026 – August 2026

- Designed, prototyped, and manufactured multiple mechanical clamping boots using SolidWorks CAD and 3D printing; optimized the final design to improve tensile load capacity by 1,300% while maintaining electrical insulation.
- Authored test reports defining a mechanical and electrical validation testing plan (tensile, compression, cantilever, and wet/dry flashover testing), performing root-cause investigation and material inspection.
- Gained hands-on exposure to automated material handling, 6-axis robotic arms, and injection molding by working alongside an automation engineer on manufacturing lines.

Landscape Laborer, O'Neill Landscaping – Elliot, ME

May 2024 – August 2025

- Promoted after one season to independently operate company vehicles and complete customer job sites based on demonstrated reliability, accountability, and sound judgment.

Projects

Autonomous Robotic Car

- Built and programmed an Arduino-based autonomous 4WD robotic vehicle integrating ultrasonic sensors to perform obstacle avoidance, line following, and PID-based navigation.
- Programmed line-following and obstacle avoidance in Arduino IDE using C++; debugged wiring/sensor-mapping issues using a circuit schematic, leading to a 15% decrease in complex line-following time.

V6 Car Engine Design & Simulation

- Modeled 50+ individual components and assembled a fully constrained V6 engine in SolidWorks.
- Created a motion study animating piston-crankshaft interaction and cylinder firing sequence.

Drone Design & Analysis

- Performed FEA on a quadcopter frame (SpaceClaim & ANSYS) under realistic loads such as thrust and gravity.
- Identified safety factors of 15 across the frame (propeller-to-hub junction as the critical stress point, 1.5 MPa, safety factor >10), revealing the design was significantly over-built and guiding where future lightweighting should focus.

Skills

Software: SolidWorks, ANSYS, SpaceClaim, MoldFlow, Simulink/LabVIEW, Visual Studio

Programming: MATLAB, Python, C++, HTML (with CSS & JavaScript), Microsoft Excel

Hardware: Arduino, Raspberry Pi, Oscilloscope, Multimeter, Soldering

Manufacturing: Additive manufacturing/3D Printing, Lean Principles, Injection Molding